Hazard Analysis
Keystone

Team 13

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Table 1: Revision History

Date	**Developer(s)**	**Change**
Oct 18th	Ayush Patel	Introduction, Scope, System Boundary and Assumptions
Oct 19th	Ahmed Rashrash	FMEA Table, Safety and Security Requirements
Oct 19th	Ayush Patel	Roadmap for Safety Req Implementation

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1 Introduction

Keystone is a capstone project discovery and management platform intended to support the lifecycle of a capstone project, from project proposal and discovery through team formation, project approval, and the management of project information. The platform is intended to serve students, project supervisors, course instructors, and teaching assistants (TAs).

Although Keystone is not intended to control physical equipment or perform safety-critical operations, failures can still cause harm. Examples include unauthorized disclosure of confidential project and student information, invalid team assignments, lost or inconsistent project updates, and misleading approval or notification status. These failures could delay academic activities, create unnecessary work, and damage trust.

We define a hazard as an unsafe system condition that, in combination with particular environmental circumstances, can contribute to an undesirable outcome. Hazard analysis examines how system failures and interactions can produce such conditions and identifies measures to eliminate or control them (**??**).

This document identifies potential hazards associated with Keystone, analyzes the ways in which system components may fail, and proposes mitigations that can be translated into verifiable safety and security requirements.

2 Scope and Purpose of Hazard Analysis

The scope includes functional failures, integration failures, security and privacy weaknesses, usability and accessibility issues, and foreseeable user errors. It also considers failures at the boundaries between Keystone and external services.

This hazard analysis covers the following categories of loss:

- **Disruption of academic activities:** Failure to discover projects, form eligible teams, obtain approvals, or maintain accurate project records may delay project selection and interfere with course milestones.
- **Unauthorized disclosure or misuse of information:** Confidential project proposals, intellectual property concerns, contact information, student profiles, and team records may be exposed to people who are not authorized to access them.
- **Incorrect academic or administrative decisions:** Inaccurate project descriptions, invalid team assignments, incorrect course rules, or misleading approval statuses may cause students, supervisors, or instructors to make decisions based on incorrect information.
- **Loss or corruption of project information:** Failed updates, integration errors, or inconsistent records may cause project details, supervisor information, repository links, or selected documentation requirements to become unavailable or outdated.

- **Loss of trust and adoption:** Unreliable notifications, unclear workflows, inaccessible interfaces, or inappropriate disclosure of information may frustrate users and reduce confidence in the platform.

3 System Boundaries and Components

3.1 System Boundary

Keystone is the system under analysis. Its boundary includes the application logic, persistent data, user-facing functionality, and integrations implemented or configured specifically for the platform. See Figure 1

External services are not assumed to be under Keystone's direct control. Nevertheless, their interfaces and the consequences of their failure are included in the analysis because Keystone depends on them to provide its intended functionality.

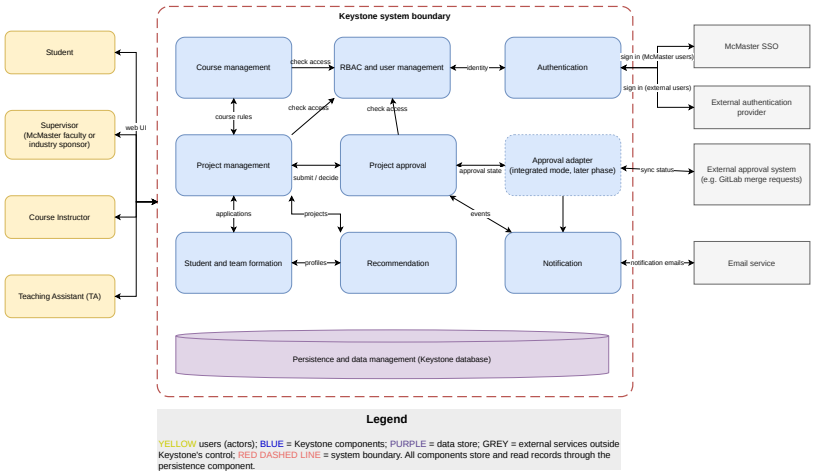


Figure 1: System and boundary context diagram for Keystone

The following external entities and services are relevant:

- **McMaster single sign-on (SSO):** Provides authentication for users accessing Keystone through their McMaster credentials.
- **External authentication provider:** Supports authentication for users who do not have a McMaster account.
- **Email service:** Delivers notifications concerning project submissions, approvals, supervisor confirmations, and other events.

- **External approval system:** A course-specific system outside Keystone where a course approves team projects, for example the GitLab merge request process used by the Software Engineering course. It is not under Keystone's control. In *external* approval mode, the instructor or a TA records the approval outcome in Keystone by hand. In *integrated* mode, an approval adapter (a Keystone component) reads the approval state through the external approval system's interface and updates Keystone's records.

3.2 Major Components

The following components define the principal areas in which failures will be investigated. They are meant to be functional boundaries for hazard identification, rather than a complete software architecture.

1. **Authentication component:** Manages user sign-in through McMaster SSO and an external authentication provider. It is responsible for establishing the identity associated with a user session. Authentication failures may allow unauthorized access or prevent legitimate users from accessing the platform.

2. **Role-Based Access Control (RBAC) and user management component:** Maintains user accounts, roles, and permissions. It determines which resources users may view or modify. Relevant roles include students, supervisors, TAs, and course instructors. Incorrect permissions may expose restricted information or allow unauthorized changes.

3. **Course management component:** Maintains course configurations, course membership, instructor and TA assignments, course-specific rules and configurations, and relevant deadlines or course information. It must accommodate differences between Software Engineering (SE), Computer Science (CS), and Mechatronics (TRON) courses.

4. **Project management component:** Manages project proposals, descriptions, supervisors, contact details, project categories, confidentiality information, intellectual property (IP) concerns, repository links, and project status throughout the project lifecycle.

5. **Project approval component:** Supports project proposal review, team project approval, and the recording of approval decisions. It must distinguish between submission, review, approval, and other workflow states so that users are not misled about whether a project or team has been approved.

6. **Student and team formation component:** Manages student profiles, team membership, project interests, and any supported team or project recommendations. It must respect course eligibility rules and team-size constraints.

7. **Notification component:** Generates and sends notifications associated with relevant workflow events via the email service. Failures may result in missing, delayed, duplicated, or incorrectly addressed messages.

8. **Persistence and data management component:** Stores and retrieves the platform's persistent records. It maintains the consistency of related information, including project records, team membership, course configuration, approval states, and user permissions.

9. **Approval adapter (integrated mode):** Reads approval state from an external approval system and updates Keystone's records. Failures may leave Keystone showing a state that differs from the external system.

10. **Recommendation component:** Ranks or filters projects and potential team matches based on available information and user preferences. Incorrect recommendations may hide relevant opportunities or mislead users about their suitability for a project.

4 Critical Assumptions

1. **Instructor authority:** Course instructors remain responsible for final project approval, the interpretation of course rules, and exceptional team-formation decisions. Keystone supports these decisions but does not replace the instructor's judgment regarding project feasibility, scope, or academic suitability.

2. **Authentication provider availability:** McMaster SSO and any external authentication provider are external dependencies and may become unavailable. Keystone cannot guarantee their continuous operation but must handle authentication failures safely.

3. **Accuracy of user-supplied information:** Students and supervisors may submit incomplete, incorrect, misleading, or outdated information. The platform can validate information according to defined rules, but it cannot guarantee the truth of every project description, supervisor commitment, or user-provided statement.

4. **Feature configuration:** Optional features may be disabled by course instructors to support different capstone courses or future deployments. Disabling a feature must not leave dependent workflows in an invalid state or expose information that would otherwise be restricted.

5 Failure Mode and Effect Analysis

Table 2: Failure Mode and Effect Analysis

Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
Authentication	SSO or external provider unavailable	Users cannot access platform, leads to missed deadlines	External service outage, network failure	Display error, log failure, raise warning to IT department and/or instructors	**SSR-1**	H1-1
	User session is stolen	Attacker can act as the user, read sensitive data, perform unapproved actions	Compromised device, unencrypted user traffic, compromised external service	Re-verify user for important actions	**SSR-9**	H1-2
RBAC and user management	User is assigned the wrong role	Student able to act as a course instructor, supervisor as an IT technician, etc.	System inferring roles from an unreliable source	Default to zero privileges if automatic assignment fails, provide instructor with ability to modify roles, and record all actions in an audit log	**SSR-2**, **SSR-10**	H2-1

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
Course management	Disabling a feature leaves dependent workflows in an invalid state	Other components fail unnecessarily and system rejects normal actions	Component state does not account for all possible configurations	Reject disable request, advise user on actions needed before feature can be disabled	**SSR-3**, **SSR-12**	H3-1
Project management	Project displays outdated information	Teams operate based on false information, contacting individuals which may no longer be involved with the course	Instructor, supervisor, or even students forget to update their information	Display a timestamp for when any displayed records were last updated	**SSR-4**	H4-1
	Confidential project or IP exposed	Potential monetary loss, loss of trust, exposed vulnerabilities	Project visibility rules not being enforced across a project	Ensure that visibility restrictions are enforced	**SSR-13**	H4-2
	Project description misrepresents project	Team cannot move forward with project, time wasted	Project proposal prompt not asking for critical information	Ensure project proposal prompt includes required fields on critical information	**SSR-14**	H4-3

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
Project approval	Approving a team for a project fails to inform other interested teams that the project is closed	Teams believe the project is still open and wait for a response that will never arrive	Component fails after completing the first approval and before updating rejected teams of new project status	Ensure all compound actions are atomic, potentially by using a database which supports atomic transactions and making the database the single source of truth	**SSR-8**	H5-1
Student and team formation	Two students attempt to join the last slot of a group simultaneously, neither receives a fail message	Both students believe they have joined the group, team size limit may be exceeded due to the data race	Database allows both operations to run concurrently without a locking mechanism	Ensure that data entries that are part of a single operation are not modified externally for the total duration of the operation	**SSR-11**	H6-1
Student and team formation	Decline or rejection is communicated bluntly, or students feel publicly ranked	Stress, feeling unfairly treated, students stop using the platform	Free-text decline messages, visible statuses with no guidance	Use respectful message templates with an optional reason, and show no public ranking of students or teams	**SSR-15**	H6-2

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
Notification	Important notification is not delivered	User misses important deadline, fails to respond to time sensitive matter	Notification component fails in the process of sending a notification, queued notification is lost	Ensure that queued notifications remain available until delivery is verified to have completed	**SSR-5**	H7-1
Persistence and data management	Data is lost or corrupted	User data is lost, system fails to operate as expected, audit log lost	Hardware degradation, accidental deletion, migration failures	Regular, automated backups which are verified to be able to restore platform to a previous, working state	**SSR-6**	H8-1
Personal data kept longer than needed	Privacy breach, loss of trust, legal exposure	No retention policy, deletion not implemented	Define a retention period and delete personal data on request and at the end of the course cycle	**SSR-16**	H8-2	

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
Recommendation	Irrelevant projects ranked highly	Users miss projects that interest them, due to the assumption that lower projects are not worth looking into	Recommendation algorithm performs badly, inputs are outdated	Provide multiple ways to sort project list	**SSR-7**	H9-1

6 Safety and Security Requirements

SSR-1 The system shall log all errors caused by external providers and notify university IT professionals and/or course instructors when they are detected.
Rationale: Outages caused by external providers prevent users from using the platform and require immediate attention to restore in a timely manner.
Traces To: FR31

SSR-2 The system shall record all user actions with the time, a user identifier, and their privilege level at the time of the action, in an append-only audit log.
Rationale: Accounts may get compromised and any actions taken may need to be reviewed and reversed.
Traces To: NFR9, FR29, FR30

SSR-3 The system shall reject all component disable requests if there are pending actions within the component.
Rationale: Disabling features may potentially leave the system in a partially invalid state. For example, disabling the project approvals component strands pending project approval requests.
Traces To: FR32

SSR-4 The system shall display a timestamp for any displayed records when they were last updated.
Rationale: Providing users with a timestamp can allow them to judge whether the information they are reading is up-to-date and can contact instructors or supervisors when that is not the case.
Traces To: FR33

SSR-5 All pending notifications shall be stored durably until delivery is verified.
Rationale: Lost notifications mean that users are not alerted of potentially time-sensitive matters and this can result in missed deadlines and loss of time.
Traces To: FR34

SSR-6 The platform shall provide a mechanism for automatically creating regular backups which can restore the platform to a previous state.
Rationale: Data loss incidents are not uncommon and can be caused by faulty hardware, accidental changes, and many others. Storing a set of backups ensures that any data loss is minimized to a backup period timespan.
Traces To: NFR12

SSR-7 The platform shall provide multiple ways to find, filter, and browse the list of projects.
Rationale: While a recommendation engine is proposed, it may be flawed

or operate on outdated data. To ensure users don't miss a project that they may favour, providing multiple ways to display the list can bring different projects to the top.
Traces To: FR5

SSR-8 All system actions which involve multiple dependent reads and writes, with at least one write, shall be atomic.
Rationale: System actions which involve multiple steps may fail at any point, leaving the system in a partially corrupt state. Atomicity guarantees that an operation either fully completes, or fully fails. Operations which fail can then simply be retried.
Traces To: NFR13

SSR-9 The system shall re-verify users for critical, hard-to-reverse actions, such as, but not limited to, updating credentials, any deletion, updating privileges, accepting a project offer, leaving a team, and other final or destructive actions. Re-verification must include at least one of: re-entering credentials, entering an email code, responding to an MFA challenge, and/or responding to an SSO challenge.
Rationale: Sessions may get stolen or unsecured devices may get compromised. Actions which are difficult to undo should be protected to ensure the actual account holder is performing the action.
Traces To: FR35

SSR-10 The system shall assign new users zero privileges when their identity cannot be assigned automatically or verified.
Rationale: No assumptions can be made about new users. They may be students who are not enrolled in a capstone course, malicious actors attempting to extract user information, individuals posing as supervisors to get free labour, etc. Ensuring no unverified users get access to the platform mitigates those risks.
Traces To: FR28, FR29, FR30, NFR5

SSR-11 The system shall ensure that operations on sets or aggregations of multiple data entries do not operate on stale data at any point during an operation.
Rationale: Concurrent operations which rely on aggregations of data entries may introduce a data race, resulting in an invalid state after the operation. For example, two students attempting to join the same team which only has one available slot can end up exceeding the team size limit, if the mechanism is not implemented correctly.
Traces To: NFR13

SSR-12 The system shall notify the user of pending actions when disabling a component fails.
Rationale: It may be possible to disable certain components but require additional actions to be taken, such as responding to pending project approvals.
Traces To: FR32

SSR-13 The system shall enforce visibility restrictions across all points of communication.
Rationale: Exposing IP or confidential information can lead to damages and loss of trust.
Traces To: FR29

SSR-14 The system shall display project proposal prompts which cover all necessary information relevant to a project's viability as a capstone project.
Rationale: Prompting only an open-ended description provides supervisors with no information on what students and instructors are looking for in a capstone-level project.
Traces To: FR1, FR2

SSR-15 The system shall present decline and rejection decisions with respectful message templates and shall not publish rankings of students or teams.
Rationale: Blunt or public decisions cause stress and a sense of unfair treatment, and may stop students from using the platform.
Traces To: FR14, FR18

SSR-16 The system shall delete personal data on request and at the end of a course cycle, and shall collect only the data that a feature needs.
Rationale: Keeping personal data longer than needed increases the harm of a breach.
Traces To: NFR11

7 Roadmap

The safety and security requirements follow the phases of the development plan in the SRS. A requirement is implemented in the same phase as the feature it protects, so that no feature is released without the protection that its hazards call for. Requirements that protect features outside the course timeline, or that depend on parts of the system that are faked for the demo, are future work.

7.1 Phase 1: Core Workflow (implemented in the course)

Phase 1 covers proposal submission and approval, browsing projects with a visible status and team selection, team project submission and review, and the validated team composition export.

- **SSR-14**: the proposal template asks for the information that makes a project viable as a capstone project (H4-3).
- **SSR-13**: server-side visibility checks for proposals and projects, tested for each role (H4-2).
- **SSR-10**: new accounts have no privileges until their role is assigned (H2-1).

- **SSR-2**: an append-only audit log for approval decisions, status changes, and role changes (H2-1). Logging every user action is future work.
- **SSR-4**: last-updated time shown on project records (H4-1).
- **SSR-7**: several ways to sort and filter the project list when browsing (H9-1). Ranked recommendations themselves are future work.
- **SSR-8**, **SSR-11**: database transactions and locking for team selection, approvals, and team membership (H5-1, H6-1).
- **SSR-1**: logging of errors from external providers (H1-1). Notifying instructors by email is added with notifications in Phase 2.
- **SSR-6**: automated database backups by the hosting provider, and a restore test before release (H8-1).
- **SSR-15**: decline and rejection message templates, with no public rankings (H6-2).

7.2 Phase 2: Communication and Visibility (implemented in the course)

Phase 2 adds notifications and the checklist, team profiles and contact details, the course schedule, and identifying unselected projects.

- **SSR-5**: pending notifications are stored in the database until delivery is confirmed, with retries (H7-1).
- **SSR-1**: alerts to the instructor when an external provider fails (H1-1).
- **SSR-13**: visibility checks are extended to team profiles, contact details, and unselected projects (H4-2).

7.3 Phase 3: Extended Features (implemented if time allows)

- **SSR-14**: the proposal prompts are extended to student-initiated proposals, with the guidelines and examples that this phase adds (H4-3).
- **SSR-8**, **SSR-11**: the same locking and transactions are applied to supervisor capacity limits (H5-1, H6-1).

7.4 Future Work (not implemented, or faked for the demo)

- **SSR-3**, **SSR-12**: rules for disabling components apply once features can be switched on and off for each course (modularity and other courses). Until then the hazard (H3-1) cannot occur.

- **SSR-9**: re-verification for critical actions with email codes or MFA depends on real authentication (McMaster accounts and external accounts), which is simplified for the demo. In the course the system only asks for confirmation before destructive actions.
- **SSR-7**: ranking and explaining recommendations, with matching of teams to projects (H9-1).
- **SSR-2**: full action logging, and tamper-evident logs.
- **SSR-1**: notifying university IT of provider outages.
- **SSR-16**: deletion on request and at the end of a course cycle (H8-2).
- Hazards of any integration with an external approval system (stale or failed synchronization) need requirements of their own when it is built.

8 Appendix — Generative AI Use Disclosure

Used Claude Sonnet 5.5 for the following,

- Asked it what are the common security risks apps face that collect profile data to come up with some ideas and reinforce some of my own.
- Used it to get more research into how data breaches can happen as well as data leaks.
- To draft a latex template for the FMEA table.
- Used it to check both documents against the checklist and rubric and asked it to flag anything that was missing and any grammar issues

Verification of AI-Generated Content

The team reviewed every AI-assisted passage before including it. References were checked against the publisher or standards body listing. The FMEA rows, requirements and roadmap were checked by the team against the SRS, including each FR and NFR number in the "Traces To" lines. The team is responsible for the accuracy, completeness, and consistency of this document.

9 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

Waqar:

1. What went well while writing this deliverable? During this deliverable, distributing work was very straightforward as there was a lot of work, but it was separable by section. It was all work we have seen before in prior courses with documentation, so it was very intuitive to write and come up with solutions/answers.

2. What pain points did you experience during this deliverable, and how did you resolve them? During this deliverable, everyone was very busy because we had midterms and a lot of assignments due alongside this report. Sometimes it also became difficult to stay updated with everyone's parts. However, because we planned out our schedule well in advance, we were able to finish everything on time while also completing our other work. Everyone also posted updates in our team chat when they got the chance, so we all stayed in the loop.

3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about? Before this deliverable we had thought about things like security issues or our app crashing, which would cause downtime. While working on the hazards, we came up with how a bug could lead to incorrect decisions being made in the actual workflow of our app, which would be chaotic. We also came up with the idea of security regarding the profiles of each person, as we wanted to collect some information to help supervisors get a better idea of who they are working with (the students).

4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider? First, data leakage is always a concern with any software product. It drives whether users can trust the developers. If we have a leak, how we handle it and what

measures we take to prevent it show users that we want to protect them and that using the app is safe. Second, availability matters because the system handles time-sensitive information such as deadlines. We want the greatest uptime we can achieve so users do not have to worry about the app going down. Lastly, data correctness is important: while we are handling PII, we want to process it correctly and avoid sharing it with third parties without user consent.

Ayush:

1. What went well while writing this deliverable?

   A great part about this document is it really made us think about our project from a different perspective. It made us think about all the ways things could go wrong, maybe an external service failed, maybe a user did something wrong. It made us think about making our design more robust so it can handle external failures. It also made us think about internal system failures that could have drastic impacts on functionality, such as RBAC not working properly and data integrity concerns. Overall, it was actually quite intuitive to write this deliverable.

2. What pain points did you experience during this deliverable, and how did you resolve them? The main issues with this deliverable were related to keeping everything consistent with the hazard analysis and the SRS. The scope of the system outlined in the SRS dictates the scope of the HA. The hazards and mitigation strategies outlined in the FMEA table must map directly to either functional, non-functional or data requirements in the SRS. But because everything is constantly changing while we are writing it, it was quite difficult to make sure both documents aligned.

3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?

   Going in, we mostly had the obvious software risks in our heads. We knew that login and permissions (SSO, external accounts, who can see what) could go wrong. We also worried about privacy via student profiles leaking, and supervisors' or students' project ideas getting "stolen". And we knew that if the platform broke, team formation and approvals would get delayed.

   Most of the more specific risks only showed up once we went component by component through the FMEA. For example, we only thought about hurt feelings and fairness when we went through the HA slides and checklist. Additionally, we had not thought about data retention policies until we properly analyzed the persistence module.

4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

One risk is security and privacy risk. This covers things like someone getting into an account they shouldn't, or personal info and confidential project ideas leaking. It matters because people trust the platform with their data, and once that trust is gone they stop using it. There can also be legal consequences, since software systems have to follow privacy law.

Psychological and social risk is a big risk in software systems. This is stress, embarrassment, feeling unfairly treated or excluded by how the software behaves. It matters because software shapes how people feel about a process. Something like a blunt rejection message or a visible ranking can really hurt someone, even though nothing "broke".